SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Solution Requirements

Date	4 July 2026
Team ID	xxxxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely access the application to use crop prediction services (login can be added if required).	High
2	Authorization levels	The system provides access to users (farmers) to enter data and view crop recommendations. The administrator can manage the dataset and trained model.	Medium

3	External interfaces	The system interacts with the crop dataset, trained machine learning model, and Flask web interface.	High
4	Transactions processing	The application accepts soil and environmental parameters (N, P, K, temperature, humidity, pH, rainfall), processes them, and generates a crop recommendation.	High
5	Reporting	Displays the predicted crop on the web page and allows analysis of model accuracy and performance.	Medium
6	Business rules, etc.	Crop recommendations are generated only after validating all input parameters. Predictions are based on the trained machine learning model.	High
7	Compliance to laws or regulations	User information is handled securely and only agricultural data is processed. The system follows general data privacy and ethical AI practices.	Medium
8	*Other*	The system should provide accurate crop recommendations with a simple and user-friendly interface.	High

Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	The application should process user input and display crop recommendations quickly.	Prediction generated in **less than 2 seconds**.
2	Scalability	The system should support multiple users and allow future expansion with additional crops and datasets.	Supports **500+ users** without performance degradation.
3	Security & Data Privacy	Protect user input and prevent unauthorized access to the system.	Secure data handling and input validation.
4	Reliability & Availability	The application should provide consistent and reliable crop recommendations.	**99% system availability** during usage.
5	Usability & Accessibility	The interface should be simple, responsive, and easy for farmers to use.	User-friendly interface with clear input fields and results.
6	*Other*	The application should be easy to update, maintain, and deploy on Windows, Linux, or macOS.	Modular Python code with Flask deployment support.